



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

C PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Programming Languages

TOTAL: 10 QUESTIONS

Q1 What are the primary design goals of C?

Threat Model

Q6 How does C implement object-oriented or functional programming?

Observability

Q2 How does C handle type checking: static or dynamic?

Architecture

Q7 How does C handle errors?

Lifecycle

Q3 How does C manage memory?

Configuration

Q8 What testing tools are commonly used in C?

Compliance

Q4 What is C's concurrency model?

Hardening

Q9 What makes C well-suited for its target domain?

Testing

Q5 How are packages and dependencies managed in C?

SSO / IdP

Q10 What are common performance pitfalls in C?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 C was designed with specific priorities

Mitigation

C was designed with specific priorities around performance, safety, or developer productivity depending on its domain. These goals shape its syntax, type system, and standard library choices.

A6 C supports classes and inheritance for

Observability

C supports classes and inheritance for OOP, or first-class functions and immutability for FP, or both. Many modern C programs blend both paradigms depending on the problem.

A2 C uses its type system to

Primitives

C uses its type system to catch errors at compile time (static) or at runtime (dynamic). This affects refactoring safety, tooling support, and the kinds of bugs that reach production.

A7 C surfaces errors via exceptions, Result/Either

Automation

C surfaces errors via exceptions, Result/Either types, or error return values. The choice impacts whether errors are checked at compile time and how calling code is structured.

A3 C uses one of three approaches:

Hardening

C uses one of three approaches: manual allocation/deallocation, a garbage collector that periodically reclaims unreachable objects, or automatic reference counting. Each has different latency and throughput tradeoffs.

A8 C has a standard or widely

Governance

C has a standard or widely adopted test runner and assertion library. Tests are typically organized into unit, integration, and end-to-end layers with coverage reporting built in or via plugins.

A4 C supports concurrency through threads,

Gotchas

C supports concurrency through threads, coroutines, async/await, or an actor model. The choice determines how I/O-bound and CPU-bound workloads are scaled and how shared state is safely accessed.

A9 C excels in its domain due

Assessment

C excels in its domain due to a combination of performance characteristics, ecosystem maturity, language ergonomics, and community support that makes common tasks simple.

A5 C uses a package manager and

Federation

C uses a package manager and a manifest file to declare dependencies and version constraints. A lock file pins exact versions to ensure reproducible builds across environments.

A10 Typical performance issues in C include

Optimization

Typical performance issues in C include excessive allocations, blocking the main thread or event loop, $O(n^2)$ data structures, and missing caching. Profiling and benchmarking tools are available to diagnose them.